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### ELECTRIC MOTOR CONTROLLER

#### Abstract

An apparatus includes an electric motor having an output shaft and a stator comprising plurality of windings and including a torque sensor that is a surface acoustic wave (SAW) sensor system, an electric motor inverter drive including a gate high side driver and a gate low side driver for each respective winding, and an electric motor controller that includes a microcontroller and a plurality of pulse width modulators (PWMs), one PWM for each gate high side and low side driver, the microcontroller being programmed to generate PWM signals for the gate drivers that represent requested torque. The output of the SAW sensor system is compared by the microcontroller with the requested torque, and the PWM signals for the gate drivers are modified by the microcontroller according to a difference between the programmed PWM signals (representing requested torque) and the electrical output signal of the SAW sensor system (representing actual torque).

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] The present application claims priority to United Kingdom Patent Application No. 2402948.0 filed on Feb. 29, 2024, the contents of which are incorporated by reference herein in their entirety.

### TECHNICAL FIELD

[0002] The present disclosure relates to electric motors and in particular to controllers for the drive motors in electric vehicles.

### BACKGROUND

[0003] Electric vehicles typically use field oriented control (FOC), to control the electric motor or motors used to propel the vehicle. FOC is preferred to other forms of electric motor control because FOC provides the smoothest and most accurate motor control. However, FOC requires large amounts of processing power due to its complexity. FOC works by mathematically transforming the motor's electrical equations into a rotating reference frame with a d and a q axis 90 degrees out of phase. The motor's torque is controlled by the d-axis current while the motor's speed is controlled by the q-axis current. The controller calculates the d-axis and q-axis currents using the motor's speed, torque and other parameters. Control of the electric motor is achieved by switching circuits in the motor inverter. Not only is the processing requirement of FOC high, but the speed and position information about the rotor must be very accurate. It is therefore necessary to use a particular type of sensor known as a resolver.

[0004] Whilst FOC provides the smoothest and most accurate control, maximum switching speeds are limited. Switching speeds are limited by the time lag between obtaining data from the resolver and reading that data into the controller.

[0005] Another known method of controlling an electric motor is known as direct torque control (DTC). Whilst DTC is a much simpler control arrangement than FOC, DTC is not generally used in traction control applications because estimation errors used in DTC cause torque ripple. This torque ripple results in an unsatisfactory level of vehicle control. Acceleration of the vehicle will not be smooth as the motor output torque is not smooth. Crucially, this can lead to unintended vehicle acceleration, driveline vibration and noise. Torque ripple has been considered an inherent limitation of direct torque control. DTC is typically only used in applications where very accurate control of motor torque and speed is not required.

[0006] As electric vehicles develop there is an increasing demand for electric motors capable of running at higher speeds, as increasing motor speed increases power density. There is also a demand for the cost of electric motors and their control systems to fall.

[0007] The limiting factor for electric motors using FOC is the time taken for processing the calculations and resolver sensor inputs necessary for FOC.

[0008] It would be desirable to provide a controller for an electric motor suitable for use in an electric vehicle in which these calculations and signal processing can occur more rapidly.

[0009] It has been discovered that by providing an electric motor with a torque sensor configured to sense the torque output of the electric motor, improved control of the electric motor can be obtained.

[0010] A surface acoustic wave sensor (SAW) is a known type of sensor that is capable of sensing torque. The Applicant's United Kingdom patent number GB2397379 describes a SAW sensor.

## SUMMARY

[0011] According to a first aspect of the present disclosure there is provided apparatus comprising an electric motor, an electric motor inverter drive and an electric motor controller, wherein: [0012] the electric motor comprises a rotor having an output shaft and a stator comprising plurality of windings; [0013] the inverter drive comprises a gate high side driver and a gate low side driver for each respective winding; [0014] the controller comprises a microcontroller and a plurality of pulse width modulators (PWMs), one PWM for each gate high side driver and each gate low side driver, the microcontroller being programmed to generate pulse width modulated signals for the gate drivers, the programmed pulse width modulated signals being representative of requested torque; and [0015] the microcontroller is connectable to a communication network; [0016] characterized in that the electric motor includes a torque sensor, the torque sensor being a surface acoustic wave (SAW) sensor system, one part of the SAW sensor system being mounted on the rotor and another part on the stator, an electrical output signal of the SAW sensor system that is representative of actual electric motor torque forming an input, to the microcontroller and wherein the electrical output signal of the SAW sensor system is compared by the microcontroller with the requested torque and the pulse width modulated signals for the gate drivers are modified by the microcontroller according to the difference between the programmed PWM signals representative of requested torque and the actual electric motor torque, represented by the electrical output signal of the SAW sensor system.

[0017] By sensing actual motor torque and using this to modify the PWM signals to the gate drivers the torque estimation calculations of the prior art control systems are no longer required. Measuring actual torque removes the need for look-up table based calculations used to generate torque estimations.

[0018] The torque sensor may be arranged to sense torque in the output shaft. The part of the SAW sensor system mounted on the rotor may be mounted on the output shaft.

[0019] The electric motor may be a three phase motor and the plurality of windings comprise three phase windings.

[0020] The apparatus may further comprise a sensor generating a signal indicative of motor speed and/or a motor temperature sensor and wherein the outputs of the respective motor speed sensor and/or motor temperature sensor are connected, and form inputs, to the microcontroller and wherein the microcontroller is programmed to generate pulse width modulated switching signals for the gate drivers based on sensed speed and/or sensed temperature.

[0021] The temperature sensor may be configured to sense temperature in the stator.

[0022] The temperature sensor may be configured to sense temperature in the stator winding.

[0023] The apparatus may further comprise a phase current sensor having an output, wherein the phase current sensor output is connected, and forms an input, to the microcontroller.

[0024] The microcontroller may be configured to generate a field weakening current signal based on the phase current sensor output.

[0025] The communication network may be a CANbus.

[0026] The part of the SAW sensor system mounted on the rotor may be a rotor coupler and the other part mounted on the stator may be a stator coupler, each coupler including a near field antenna.

[0027] The apparatus may further comprise at least one sensor connected electrically to the near field antenna of the rotor coupler.

[0028] The rotor may include at least one flat surface, the at least one sensor mounted on the at least one flat surface.

[0029] The rotor may include two flat surfaces separated by 180 degrees, wherein each of the flat surfaces has one of the at least one sensors mounted thereon.

[0030] The near field antenna of the stator coupler may be connected electrically to an interrogation unit.

[0031] The interrogation unit may include an application specific integrated circuit (ASIC).  
[0032] The interrogation unit, for example the ASIC thereof, may generate a radio frequency interrogation signal having a frequency range. The frequency range may be 420-440 MHz.  
[0033] One of the sensors may be configured to pick up the radio frequency signal in a first part of the frequency range and the other of the sensors may be configured to pick up the radio frequency signal in a second part of the frequency range. The first part of the frequency range may be 420 to 430 MHz and the second part of the frequency range may be 430-440 MHz.  
[0034] A back scattered signal from the at least one sensor may be reflected back to the interrogation unit, for example the ASIC thereof, by the near field antennas and may be read by the interrogation unit, for example by the ASIC.  
[0035] An output signal from interrogation unit, typically the ASIC, may form an input to the microcontroller, thus enabling control of the motor according to the torque sensed in the motor, typically in the output shaft of the motor.  
[0036] According to a second aspect of the invention there is provided an electric vehicle comprising at least one apparatus of the first aspect of the invention.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0037] In the Drawings, which illustrate a prior art example of a motor and associated control arrangement and the motor and associated control arrangement of the present disclosure, which is by way of example:  
[0038] FIG. 1 is a schematic representation of an example of an FOC motor control arrangement of the prior art;  
[0039] FIG. 2 is a schematic representation of an example of a motor control arrangement of the present disclosure;  
[0040] FIG. 3 is a schematic representation of a motor of the present disclosure;  
[0041] FIG. 4a is a schematic representation of the assembled motor illustrated FIG. 3;  
[0042] FIG. 4b is an exploded view of the motor illustrated in FIG. 3; and  
[0043] FIG. 5 is an exploded view of the SAW sensor assembly of the motor illustrated in FIGS. 3 to 4b.

### DETAILED DESCRIPTION

[0044] Referring first to FIG. 1, a three phase electric motor 1 comprises three sets of phase windings, phase A, phase B and phase C.  
[0045] The motor 1 is driven by an inverter 2, which comprises respective high side gate drivers GHA, GHB and GHC and low side gate drivers GLA, GLB and GLC.  
[0046] An input voltage (B+ B-) is applied across the inverter and the input current is sensed by current sensor CS1.  
[0047] Each set of windings is provided with a current sensor, current sensor A (CSA), current sensor B (CSB) and current sensor C (CSC), the respective current sensors sensing the current in the winding A-C to which the current sensor CSA to CSC is assigned.  
[0048] An inverter controller 3 comprises a microcontroller 4 and two sets of three pulse width modulators 5a, 5b. Each pulse width modulator of the set of three pulse width modulators 5a is connected to a respective one of the gate high side drivers GHA, GHB and GHC. Similarly, each pulse width modulator of the set of three pulse width modulators 5b is connected to a respective one of the gate low side drivers GLA, GLB and GLC.  
[0049] The microcontroller 4 is also connected to an external communications network, such as a CANbus of a vehicle of which the motor 1 forms a part. The current sensors CS1, CSA, CSB and CSC are also connected to the microcontroller 4, providing inputs thereto.

[0050] The electric motor **1** is provided with a temperature sensor in the form of a thermistor that is arranged to sense the temperature of one or more of the phase windings A to C. The output of the thermistor is connected to the microcontroller **4** as an input thereto.

[0051] The electric motor **1** is also provided with a sinusoidal encoder which provides sine and cosine signals as the rotor (not shown) of the motor **1** rotates. The sine and cosine signals give an indication of rotor position at any moment. The sine and cosine signals are provided as inputs **9** to the microcontroller **4** and are used to calculate motor speed and estimate the position of the rotor.

[0052] In use, the speed/torque demand is transmitted to the microcontroller **4** by the CANbus **6** when a driver of a vehicle of which the motor **1** forms a part, gives a command, for example by pressing on an accelerator pedal.

[0053] The sine and cosine signals from the encoder are provided to the microcontroller **4**. The microcontroller **4** calculates motor speed and estimates the position of the rotor.

[0054] The currents sensed by CS1, CSA, CSB and CSC are provided as current sense inputs **7** to the microcontroller **4**. The microcontroller **4** is programmed with an algorithm, such as a Clarkes transform or a Park transform through which a torque estimation is derived from the inputs to the microcontroller **4**.

[0055] The microcontroller **4** calculates the pulse width modulated switching for the gate drivers GHA, GHB, GHC and GLA, GLB and GLC that provide for the correct regulation of current flowing to and from the motor **1** to achieve a desired output shaft speed and torque.

[0056] Where additional field weakening is required the microcontroller must also be able to calculate and create a suitable secondary stator current flow.

[0057] In FIGS. **1** and **2**, like numerals are used to indicate like parts.

[0058] FIG. **2** illustrates a motor and motor control arrangement of the invention having direct torque control with active torque feedback to the inverter, which comprises a three phase electric motor **1** having three sets of phase windings, phase A, phase B and phase C.

[0059] The motor **1** is driven by an inverter **2**, which comprises respective high side gate drivers (GHA, GHB and GHC) and low side gate drivers GLA, GLB and GLC.

[0060] An input voltage (B+ B-) is applied across the inverter and the input current is sensed by current sensor CS1.

[0061] Whilst the phase A set of windings is shown as being provided with a current sensor A (CSA), the current sensor CSA is optional.

[0062] An inverter controller **3** comprises a microcontroller **4** and two sets of three pulse width modulators **5a**, **5b**. Each pulse width modulator of the set of three pulse width modulators **5a** is connected to a respective one of the gate high side drivers GHA, GHB and GHC. Similarly, each pulse width modulator of the set of three pulse width modulators **5b** is connected to a respective one of the gate low side drivers GLA, GLB and GLC.

[0063] The microcontroller **4** is also connected to an external communications network, such as a CANbus of a vehicle of which the motor **1** forms a part. The current sensors CS1 and if present CSA are also connected to the microcontroller **4**, providing inputs thereto.

[0064] The electric motor **1** is provided with at least one temperature sensor, in the form of a thermistor or plurality of thermistors, arranged to sense the temperature of one or more of the phase windings A to C. The output **8** of the thermistor is connected to the microcontroller **4** as an input thereto.

[0065] The electric motor **1** is also provided with a speed sensor which measures the rotational speed of the motor **1** and provides a motor speed input signal **10** to the microcontroller **4**.

[0066] The electric motor **1** is also provided with a torque sensor TS1 arranged to measure torque in the output shaft **1'** of the motor **1**. The torque sensor is a surface acoustic wave (SAW) sensor system, for example of the type described in GB2397379.

[0067] In use, the speed/torque demand is transmitted to the microcontroller **4** by the CANbus **6** when a driver of a vehicle of which the motor **1** forms a part, gives a command, for example by

pressing on an accelerator pedal. The microcontroller is programmed to generate pulse width modulated signals for the gate drivers of the inverter **2**, initially according to commands, such as accelerator pedal position.

[0068] The signal **10** from the motor speed sensor is compared with the speed demand transmitted to the gate drivers of the inverter **2**.

[0069] Similarly, the signal **11** from the output shaft torque sensor TS**1** is compared with the torque demand transmitted to the gate drivers of the inverter **2**.

[0070] The microcontroller **4** calculates the differences between speed demand and measured motor speed, and torque demand and the torque measured in the output shaft **1'** and uses these calculated differences to calculate the required pulse width modulation for the gate drivers so that the current flowing to the and from the phase windings of the motor cause the motor to operate at the desired speed with the desired output shaft torque.

[0071] Where additional field weakening is required the microcontroller can also calculate and create a suitable secondary stator current flow. The phase current sensor A provides the signal that allows the field weakening current to be calculated.

[0072] Referring now to FIGS. **3** and **4a**, the motor **1** comprises a motor frame **1a**, a stator **1b**, a rotor **1c**, an output shaft **1d**, an end plate **1e**, and support bearings **1f** supporting each end of the output shaft **1d**. A surface acoustic wave (SAW) sensor assembly **20** is mounted on the output shaft **1d**.

[0073] FIGS. **4a** and **5** illustrate the SAW sensor assembly **20** in greater detail. The SAW sensor assembly **20** comprises a (PCB) stator coupler **21** which includes a collar **21a**, typically formed from glass reinforced plastic. The collar **21a** mounts an outer electrically conductive layer forming a ground plane antenna **21b** and an inner electrically conductive track forming a near field antenna **21c**. Typically, the electrically conductive components are formed from copper, although any suitable electrical conductor may be used. The SAW sensor assembly **20** further comprises a printed circuit board (PCB) rotor coupler **22**, which includes a collar **22a**, typically formed from glass reinforced plastic. The collar **22a** mounts an outer electrically conductive layer track forming a near field antenna **22b**.

[0074] The stator and rotor couplers are separated by a small distance, typically, between 1 and 20 mm.

[0075] The rotor coupler **22** is fixed, for example by an adhesive, to a rotor coupler mounting region **1d''** of the output shaft **1d**, the rotor coupler mounting region **1d''** being cylindrical in the illustrated example. The output shaft **1d** is also provided with a sensor mounting region **1d'**, the sensor mounting region **1d'** including flat surfaces **23a**, **23b** opposite each other on the shaft **1d**. A high frequency sensor **24a** is mounted on the flat surface **23a**, with a low frequency sensor **24b** being mounted on the flat surface **23b**. By mounting two sensors opposite each other, bending of the shaft can be compensated for. In the illustrated embodiment the sensors **24a**, **24b** are all quartz packaged sensors. They require no power other than that in the radio frequency signal they receive and transmit. Such sensors are known in the prior art and therefore are not described in detail.

[0076] The sensors **24a**, **24b** are connected electrically to the near field antenna **22b** of the rotor coupler **22**. The electrically conductive track forming the near field antenna **22b** is comprised of two parts with a small gap between those two parts. Each of the sensors **24a**, **24b** is connected to both parts of the electrically conductive track by pairs of wires **22c** and **22c'** respectively.

[0077] The near field antenna **21c** of the stator coupler **21** is configured for electrical connection to an interrogation unit which includes an application specific integrated circuit (ASIC). As can be seen from FIG. **5**, the collar **21a** of the stator coupler includes a hole **21d**. The end terminals of a pair of wires through the hole **21d** and attach to respective parts of the near field antenna **21c**, one terminal being connected to respective ends **21c'**, **21c''** of the near field antenna **21c**. The stator coupler **21** is mounted in a stator coupler housing **25** which includes attachment members **25a**, each including a hole **25b** that receives a screw for attaching the stator coupler housing **25** to the motor

end plate **1e**. The stator coupler housing **25** also provides an opening **25c** through which the wire attaching the near field antenna **21c** to the interrogation unit may be passed.

[0078] In use, the ASIC generates a radio frequency interrogation signal having frequency of 420 to 440 MHz in this example. This signal is transmitted to the stator near field antenna **21c** and from the stator near field antenna **21c** to the rotor near field antenna **22b**. This radio frequency signal is transmitted from the rotor near field antenna **22b** to the sensors **24a**, **24b** by the electrical connections **22c**, the sensor **24a** picking up the signal in the range 430 to 440 MHz and the sensor **24b** picking up the signal in the range 420 to 430 MHz. The sensors **24a**, **24b** generate a back scattered signal which changes according to the torque on the shaft **1d** for example. The back scattered signal from sensors **24a**, **24b** is reflected back to the interrogation unit by the near field antennas **22b**, **21c** and is read by the interrogation unit's ASIC. An output signal from the ASIC forms an input to the microcontroller **4** illustrated in FIG. **2**, thus enabling control of the motor **1** according to the torque sensed in the output shaft **1d**.

[0079] By providing SAW sensors to sense motor speed and output shaft torque the encoder generating a sine/cosine output is not required and there is no need to perform Clarke or Park transforms to estimate torque as required by FOC electric motor control.

[0080] In the case of DTC electric motor control, the torque estimation that limits the applications where DTC electric motors can be used is replaced by measured torque, allowing more accurate control of the motor with no torque ripple.

## Claims

**1.** Apparatus comprising an electric motor, an electric motor inverter drive and an electric motor controller, wherein: the electric motor comprises a rotor having an output shaft and a stator comprising plurality of windings; the inverter drive comprises a gate high side driver and a gate low side driver for each respective winding; the controller comprises a microcontroller and a plurality of pulse width modulators (PWMs), one PWM for each gate high side driver and each gate low side driver, the microcontroller being programmed to generate pulse width modulated signals for the gate drivers, the programmed pulse width modulated signals being representative of requested torque; the microcontroller is connectable to a communication network; the electric motor includes a torque sensor, the torque sensor being a surface acoustic wave (SAW) sensor system, a first part of the SAW sensor system being mounted on the rotor and a second part of the SAW sensor system being mounted on the stator, and an electrical output signal of the SAW sensor system that is representative of actual electric motor torque forming an input to the microcontroller; and the electrical output signal of the SAW sensor system is compared by the microcontroller with the requested torque, and the pulse width modulated signals for the gate drivers are modified by the microcontroller according to a difference between the programmed PWM signals representative of requested torque and the actual electric motor torque, represented by the electrical output signal of the SAW sensor system.

**2.** Apparatus according to claim 1, wherein the torque sensor is arranged to sense torque in the output shaft.

**3.** Apparatus according to claim 1, wherein the electric motor is a three phase motor and the plurality of windings comprise three phase windings.

**4.** Apparatus according to claim 1, further including at least one of a motor speed sensor generating a signal indicative of motor speed or a motor temperature sensor, having outputs that are connected, and form inputs, to the microcontroller, wherein the microcontroller is programmed to generate pulse width modulated switching signals for the gate drivers based on sensed motor speed and/or sensed motor temperature.

**5.** Apparatus according to claim 4, wherein the motor temperature sensor is included and is configured to sense temperature in the stator.

6. Apparatus according to claim 5, wherein the motor temperature sensor is included and is configured to sense temperature in the windings of the stator.
7. Apparatus according to claim 1, further comprising a phase current sensor having an output, wherein the phase current sensor output is connected, and forms an input, to the microcontroller.
8. Apparatus according to claim 7, wherein the microcontroller is configured to generate a field weakening current signal based on the phase current sensor output.
9. Apparatus according to claim 1, wherein the communication network is a CANbus.
10. Apparatus according to claim 1, wherein the first part of the SAW sensor system mounted on the rotor is a rotor coupler and the second part of the SAW sensor system mounted on the stator is a stator coupler, the rotor coupler and the stator coupler each including a near field antenna.
11. Apparatus according to claim 10, further comprising at least one sensor connected electrically to the near field antenna of the rotor coupler.
12. Apparatus according to claim 11, wherein the rotor includes at least one flat surface, the at least one sensor mounted on the at least one flat surface.
13. Apparatus according to claim 12, wherein the at least one sensor comprises a plurality of sensors, and the rotor includes two flat surfaces separated by 180 degrees, wherein each of the flat surfaces has one of the plurality of sensors mounted thereon.
14. Apparatus according to claim 10, wherein the near field antenna of the stator coupler is connected electrically to an interrogation unit.
15. Apparatus according to claim 14, wherein the interrogation unit includes an application specific integrated circuit (ASIC).
16. Apparatus according to claim 15, wherein the ASIC generates a radio frequency interrogation signal having a frequency range.
17. Apparatus according to claim 16, wherein the at least one sensor comprises a plurality of sensors, and wherein a first one of the plurality of sensors is configured to pick up the radio frequency interrogation signal in a first part of the frequency range and a second one of the plurality of sensors is configured to pick up the radio frequency signal in a second part of the frequency range.
18. Apparatus according to claim 14, wherein a back scattered signal from the at least one sensor is reflected back to the interrogation unit by the near field antennas and is read by the interrogation unit.
19. Apparatus according to claim 18, wherein an output signal from the interrogation unit is the output of the SAW sensor system and the input to the microcontroller.
20. An electric vehicle including at least one apparatus comprising an electric motor, an electric motor inverter drive and an electric motor controller, wherein: the electric motor comprises a rotor having an output shaft and a stator comprising plurality of windings; the inverter drive comprises a gate high side driver and a gate low side driver for each respective winding; the controller comprises a microcontroller and a plurality of pulse width modulators (PWMs), one PWM for each gate high side driver and each gate low side driver, the microcontroller being programmed to generate pulse width modulated signals for the gate drivers, the programmed pulse width modulated signals being representative of requested torque; the microcontroller is connectable to a communication network; the electric motor includes a torque sensor, the torque sensor being a surface acoustic wave (SAW) sensor system, a first part of the SAW sensor system being mounted on the rotor and a second part of the SAW sensor system being mounted on the stator, and an electrical output signal of the SAW sensor system that is representative of actual electric motor torque forming an input to the microcontroller; and the electrical output signal of the SAW sensor system is compared by the microcontroller with the requested torque, and the pulse width modulated signals for the gate drivers are modified by the microcontroller according to a difference between the programmed PWM signals representative of requested torque and the actual electric motor torque, represented by the electrical output signal of the SAW sensor system.



